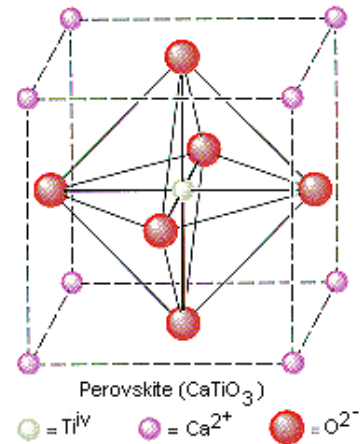
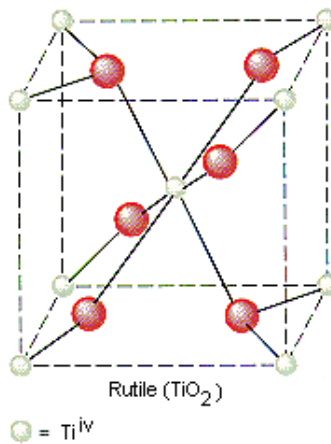
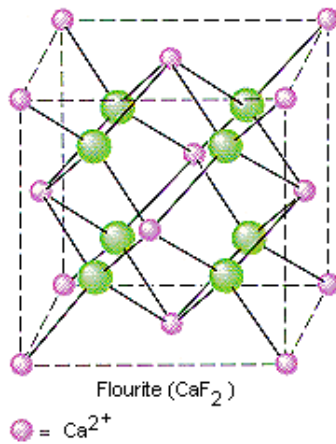
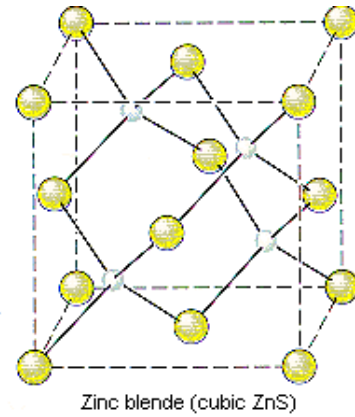
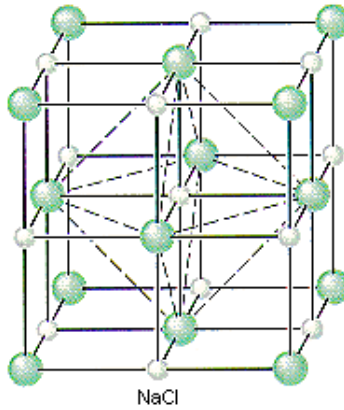
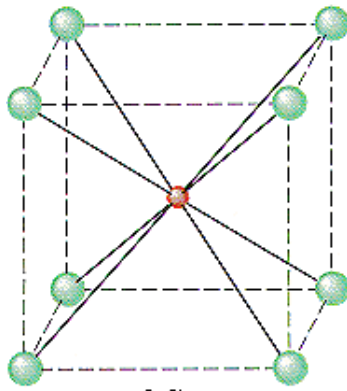


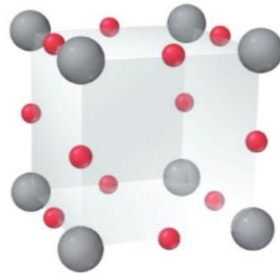
Homework 2

Deadline: Oct. 10.

- Discuss the structures of the following crystals by the method we learned, i.e. cations fill to the interstitial sites (voids) of the close packing structure of anions.

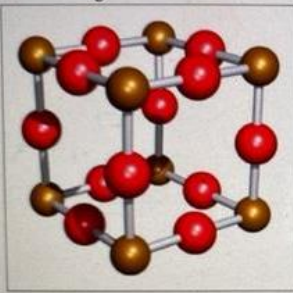


- Rhenium oxide does not have a close-packed structure. Its structure can be best described as a simple cubic array of rhenium ions with the oxide ions at the center of each edge of the cubic unit cell. What is the charge of the rhenium ion in this compound?



-

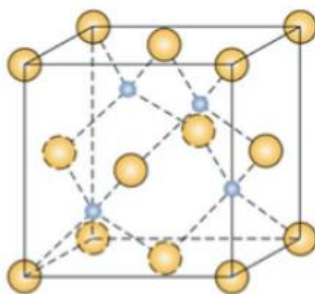
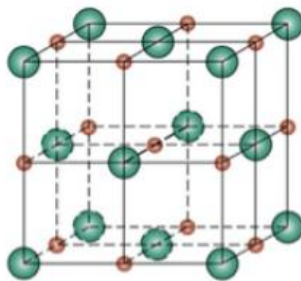
The unit cell of rhenium oxide lattice is shown below. Every stick here is a bond. Determine the following. (15 pts)



- Number of atoms of each type per unit cell.
- Formula of this compound.
- Coordination numbers of both Re and O.
- Coordination geometry of both Re and O.
- The sub-lattice type formed by Re atoms.

4.

- a) (5 pts) Identify the crystal structures below and determine the number of atoms per unit cell for each of the following unit cells:



5. List 1~2 characterization techniques to discriminate between crystalline and amorphous solid.
6. Please list the four types of bonding in solids. And give an example for each bonding and order their strengths.
7. Please list the seven basic unit cells and give the unit-cell parameters for each one (a , b , c and $\alpha\beta\gamma$)
8. The names of the two kinds of closest packing in crystals? And calculate how many particles in a primitive unit cell?
9. Please give the coordination numbers for trigonal, tetrahedral, octahedral and cubic voids.
10. Why plastic deformation occurs at stresses that are much smaller than the theoretical strength of perfect crystals?
11. Explain why the color of Fe^{2+} doped beryl crystal is blue, while the color of Fe^{3+} doped same crystal is yellow?
12. List the steps in a sol-gel process? How does the pH value affect the morphology of silica gel?